

BioSecure AI Setup & Execution Guide

A Multi-Platform Guide for Running the Shared Face Attendance System

Multi-Platform Execution Notice: This package contains pre-configured credentials (.env file). Because Gunicorn does not support Windows natively, and permission settings vary across operating systems, this guide details how to execute the application on both Linux and Windows environments.

1. Install System Dependencies

Make sure your host operating system has the correct Python configurations and runtime libraries:

```
# On Linux (Ubuntu/Debian):  
sudo apt update && sudo apt install -y python3-pip python3-venv libgl1-mesa-glx libglib2.0-0  
  
# On Windows / macOS:  
Ensure Python 3.10 or 3.11 is installed on your system PATH.
```

2. Recreate the Python Virtual Environment

Initialize a clean local virtual sandbox environment and compile packages to fit your path and system architecture:

```
# 1. Clean previous environment folder (if exists)  
rm -rf .venv # On Linux/macOS  
rmdir /s /q .venv # On Windows command prompt  
  
# 2. Initialize a fresh virtual environment  
python3 -m venv .venv # On Linux/macOS  
python -m venv .venv # On Windows  
  
# 3. Activate and Install  
source .venv/bin/activate && pip install -r requirements.txt # Linux/macOS  
.venv\Scripts\activate.bat && pip install -r requirements.txt # Windows CMD
```

3. Browser Access Instructions

The system database has been pre-configured. When running the application locally on either port 5000 or port 8000, please pay attention to the browser access instructions below:

■■ **Crucial Browser Access Note:** Gunicorn runs on a loopback interface binding to 0.0.0.0:8000. While clicking the link printed in the terminal (<http://0.0.0.0:8000>) might work on some browsers, it is highly recommended to manually navigate to **<http://localhost:8000>** or **<http://127.0.0.1:8000>** in your browser address bar to guarantee successful session cookie management and stable logins.

4. Launching on Linux / macOS

On Linux systems, startup shell scripts may lack executable execution privileges by default. Run the scripts explicitly using the bash interpreter or grant permissions first:

```
# Option A: Run directly through Bash interpreter (no permissions needed)
bash start_face_attendance.sh

# Option B: Grant executable permissions and execute
chmod +x start_face_attendance.sh
./start_face_attendance.sh

# Option C: Running development server directly (Port 5000)
flask --app app:app run --debug
```

5. Launching on Windows

Gunicorn (used in the Linux shell script) does not natively support Windows. A dedicated Windows Command batch script is provided instead:

```
# Option A: Run the pre-packaged Windows batch file
Double-click on start_face_attendance.bat (or run start_face_attendance.bat in CMD)

# Option B: Run via command line manually
flask --app app:app run --debug
```